

# ACEnet Basics Workshops

## Introduction to ACEnet

What is ACEnet and what can it do for me? How do I get an account? Once I've got an account, how do I get connected and start doing things?

|                               |                                |                             |
|-------------------------------|--------------------------------|-----------------------------|
| Dalhousie University          | Wednesday, May 1, 12:30 PM ADT | Killam basement B260        |
| Dalhousie (Sexton Campus)     | Friday, May 3, 12:30 PM ADT    | Mechanical Engineering C350 |
| St. Francis Xavier University | Monday, May 6, 2:00 PM ADT     | PSC1096                     |
| University of New Brunswick   | Wednesday, May 1, 12:30 PM ADT | GE-131                      |
| Memorial University           | Wednesday, May 1, 1:00 PM NDT  | EN-4034                     |

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## Introduction to Unix Command Line

Learn the basic commands and concepts needed to work with the Unix/Linux command line interface.

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|-------------------------------|-------------------------------|----------------------|
| Dalhousie University          | Monday, May 6, 12:30 PM ADT   | Killam basement B260 |
| St. Francis Xavier University | Wednesday, May 8, 2:00 PM ADT | PSC1096              |
| University of New Brunswick   | Monday, May 6, 12:30 PM ADT   | GE-131               |
| Memorial University           | Friday, May 3, 1:00 PM NDT    | EN-4034              |

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## Introduction to Shell Scripting

Exercise the power of the command line with shell scripting.

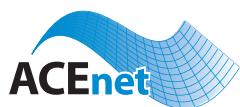
|                               |                               |                      |
|-------------------------------|-------------------------------|----------------------|
| Dalhousie University          | Monday, May 13, 12:30 PM ADT  | Killam basement B260 |
| St. Francis Xavier University | Friday, May 10, 2:00 PM ADT   | PSC1096              |
| University of New Brunswick   | Monday, May 13, 12:30 PM ADT  | GE-131               |
| Memorial University           | Wednesday, May 8, 1:00 PM NDT | EN-4034              |

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## Job Management with Grid Engine

How the job scheduler works, and how to make it work better for you.

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|-------------------------------|--------------------------------|----------------------|
| Dalhousie University          | Wednesday, May 8, 12:30 PM ADT | Killam basement B260 |
| St. Francis Xavier University | Monday, May 13, 2:00 PM ADT    | PSC1096              |
| University of New Brunswick   | Wednesday, May 8, 12:30 PM ADT | GE-131               |
| Memorial University           | Friday, May 10, 1:00 PM NDT    | EN-4034              |



# ACEnet Focus Workshops

## Perl

An introduction to the Perl scripting language. This is a local workshop at St. Francis Xavier University, but could be made available over videoconference with sufficient interest.

### Introduction to Perl

Instructor: Shah Razul

Location: PSC1096, St. FX

Wednesday, May 14, 10:00 AM

ADT, 1:00 PM NDT

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## Grid Engine

How to submit multiple jobs without typing qsub each time. Templates, loops, task arrays. Delivered over videoconference in your local ACEnet collaboration room.

### Managing Many Jobs

Instructor: Ross Dickson

Friday, May 17, 12:30 PM ADT,

1:00 PM NDT

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## Python

Get acquainted with a popular and easy to learn programming language. Delivered over videoconference in your local ACEnet collaboration room.

### Introduction to Python

Instructor: Joey Bernard

Wednesday, May 22, 12:30 PM

ADT, 1:00 PM NDT

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### Python Add-ons: SciPy & NumPy

Instructor: Joey Bernard

Friday, May 24, 12:30 PM ADT,

1:00 PM NDT

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## Parallel Programming

An introduction to parallel programming and two of the most common parallel APIs: MPI and OpenMP. Delivered over videoconference in your local ACEnet collaboration room.

### Parallel Programming Overview

Instructor: Ross Dickson

Friday, May 31, 12:30 PM ADT,

1:00 PM NDT

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### MPI: Point-to-point Communication

Instructor: Alejandro Allievi

Wednesday, June 5, 12:30 PM

ADT, 1:00 PM NDT

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### MPI: Collective Communication

Instructor: Alejandro Allievi

Friday, June 7, 12:30 PM ADT,

1:00 PM NDT

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### OpenMP: Introduction

Instructor: Joey Bernard

Wednesday, June 19, 12:30 PM

ADT, 1:00 PM NDT

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### OpenMP: Shared Memory Programming

Instructor: Joey Bernard

Friday, June 21, 12:30 PM ADT,

1:00 PM NDT

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## GROMACS

An introduction to Molecular Dynamics techniques via the popular and versatile open source software package, GROMACS. Delivered over videoconference in your local ACEnet collaboration room.

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|---------------------------------------------------------------------|-----------------------------------------------|
| <b>GROMACS I: Intro &amp; Principles</b><br>Instructor: Shah Razul  | Monday, June 10, 12:30 PM ADT,<br>1:00 PM NDT |
| <b>GROMACS II: Properties &amp; Tools</b><br>Instructor: Shah Razul | Monday, June 17, 12:30 PM ADT,<br>1:00 PM NDT |
| <b>GROMACS III: Forcefields</b><br>Instructor: Shah Razul           | Monday, June 24, 12:30 PM ADT,<br>1:00 PM NDT |

## Matlab

Learn how to solve and accelerate computationally and data-intensive problems using multicore processors, GPUs, and computer clusters.

11:00 - 12:00 Parallel & Distributed Computing  
12:00 - 1:00 Lunch courtesy of Mathworks  
1:00 - 2:00 GPU Computing with Matlab

|                                    |                                                                                                     |                                  |
|------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------|
| <b>University of New Brunswick</b> | <b>Parallel and GPU Computing with Matlab</b><br>Instructor: Mathworks Engineer<br>Location: GC127  | Tuesday, June 11, 11:00 AM ADT   |
| <b>Dalhousie University</b>        | <b>Parallel and GPU Computing with Matlab</b><br>Instructor: Mathworks Engineer<br>Location: TBA    | Wednesday, June 12, 11:00 AM ADT |
| <b>Memorial University</b>         | <b>Parallel and GPU Computing with Matlab</b><br>Instructor: Mathworks Engineer<br>Location: EN4034 | Thursday, June 13, 11:00 AM NDT  |

## R

An introduction to the statistical computing language R, a standard tool for any Big Data and Data Analytics researcher. Delivered over videoconference in your local ACEnet collaboration room.

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|------------------------------------------------------|-----------------------------------------------|
| <b>Introduction to R</b><br>Instructor: Joey Bernard | Friday, June 14, 12:30 PM ADT,<br>1:00 PM NDT |
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To ensure your local collaboration room is set-up for you, please register for any workshop you plan to attend: <http://acenet-workshops.eventbrite.ca>

### ACEnet Collaboration Rooms

| Dal                    | SMU                 | StFX                    | CBU                      | Acadia              | UPEI              | UNB                   | UNBSJ  | MUN                      |
|------------------------|---------------------|-------------------------|--------------------------|---------------------|-------------------|-----------------------|--------|--------------------------|
| Chase Bldg<br>Room 305 | Science<br>Room 127 | 42 West St.<br>Room 203 | Technology<br>Center T35 | Huggins<br>Room 154 | Duffy<br>Room 202 | Gillin Hall<br>GE 131 | IH 215 | Earth Sciences<br>ER4015 |